



IILM UNIVERSITY

Greater Noida, Uttar Pradesh

Internship Presentation

AI-Powered Automated Customer Support Chatbot with Intent Recognition and CI/CD Deployment

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PROGRAM & SEMESTER	B.Tech CSE(AI&ML), 7 th Semester
ORGANIZATION	EduSkills Academy (AICTE - National Internship Portal)

Agenda

- 1 About the Internship Program
- 2 Problem Statement & Objectives
- 3 Scope & Technology Stack
- 4 System Architecture
- 5 Methodology & Execution Timeline
- 6 CI/CD Automation Pipeline
- 7 Results & Performance Evaluation
- 8 Key Outcomes, Limitations & Future Scope

About the Internship Program

8

WEEKS

Virtual Internship Duration
June - August 2026

Grade Achieved

0 (Outstanding)

EduSkills Academy is a nation-building skills initiative that delivers virtual internships in partnership with **AICTE** (All India Council for Technical Education) and the **National Internship Portal**, an initiative supported by the Ministry of Education, Government of India.

- **AI Deployment & Automation Track**

Focused on building, containerizing and automating deployment of AI/ML applications.

- **Industry-Standard Toolchain**

Hands-on exposure to Hugging Face, PyTorch, FastAPI, Docker and GitHub Actions.

- **Mentor-Guided Execution**

Structured 8-week roadmap with weekly milestones and deliverables.

Problem Statement & Objectives

THE PROBLEM

Organizations receive a high volume of repetitive, tier-1 customer support queries (order status, refunds, account queries). Manual handling increases operational cost and response time, and pulls agents away from complex issues that truly need human judgment.

There is a need for an intelligent system that understands customer intent in natural language and can be deployed and updated reliably, without manual intervention.

PROJECT OBJECTIVES

- 1 Fine-tune DistilBERT for accurate customer-intent classification
- 2 Build a high-performance FastAPI REST endpoint for real-time inference
- 3 Containerize the application with Docker for portable deployment
- 4 Automate testing, build & deployment using GitHub Actions CI/CD
- 5 Deploy on AWS EC2 / Render for public accessibility

Scope & Technology Stack

The project covers intent-based handling of tier-1 queries — model fine-tuning, API development, containerization and CI/CD-based automated deployment. Advanced multi-turn dialogue and generative responses are identified as future scope.



AI / ML

Hugging Face + PyTorch



Validation

Pydantic Schemas



Backend API

FastAPI



Container

Docker & Compose



CI/CD

GitHub Actions



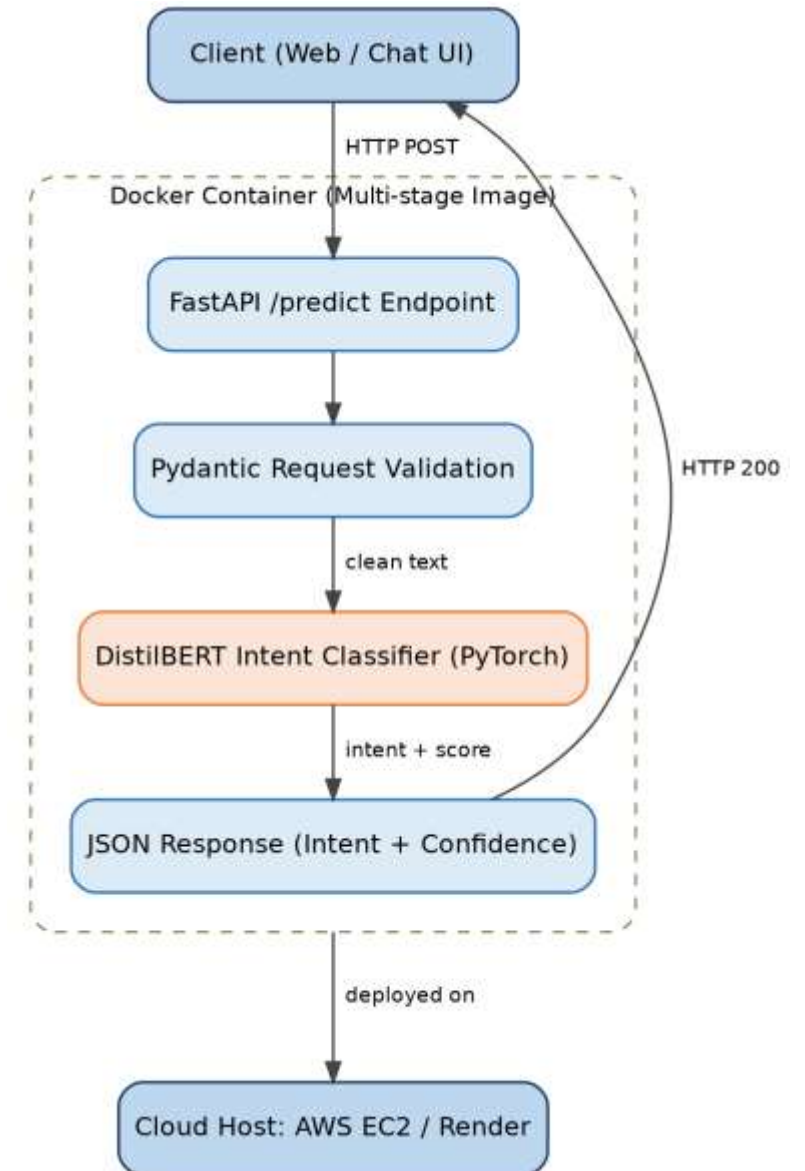
Cloud Host

AWS EC2 / Render

System Architecture

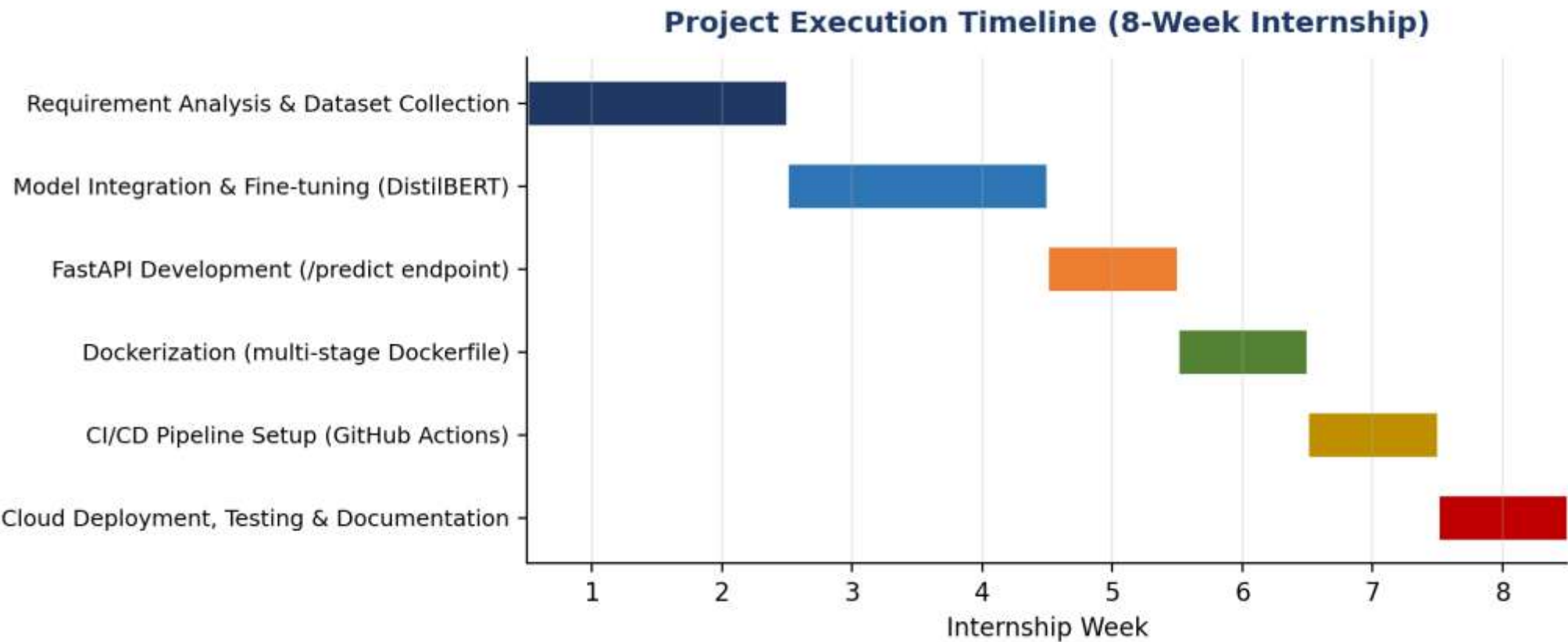
A modular, containerized microservice design: FastAPI handles routing, Pydantic validates input, and the fine-tuned DistilBERT model performs inference — all packaged in a single Docker image and deployed to the cloud.

1. Client sends request via HTTP POST
2. FastAPI /predict endpoint receives it
3. Pydantic validates the payload
4. DistilBERT classifies intent (PyTorch)
5. JSON response returned with confidence score



Methodology & Execution Timeline

The project followed a structured, phase-wise roadmap across the 8-week internship, moving from data preparation to a fully automated, cloud-deployed system.



Each phase builds on the last — from dataset preparation and model fine-tuning, through API development and containerization, to a fully automated CI/CD-driven cloud deployment by week 8.

CI/CD Automation Pipeline

A GitHub Actions workflow tests, builds and deploys the application automatically on every push to the main branch — removing manual deployment steps entirely.

1

Run Tests & Linting

Every push automatically triggers unit tests and code-quality checks.

2

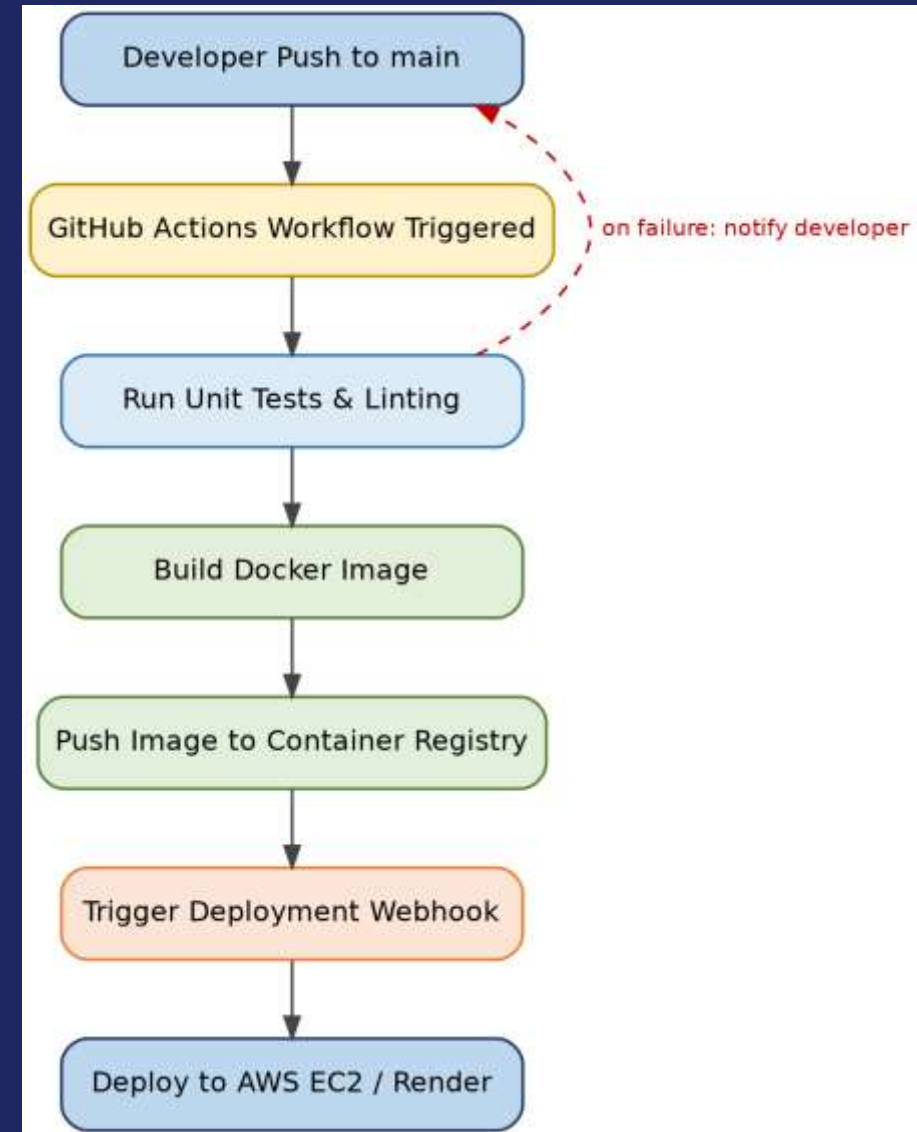
Build & Push Image

A multi-stage Dockerfile produces a minimal, secure production image.

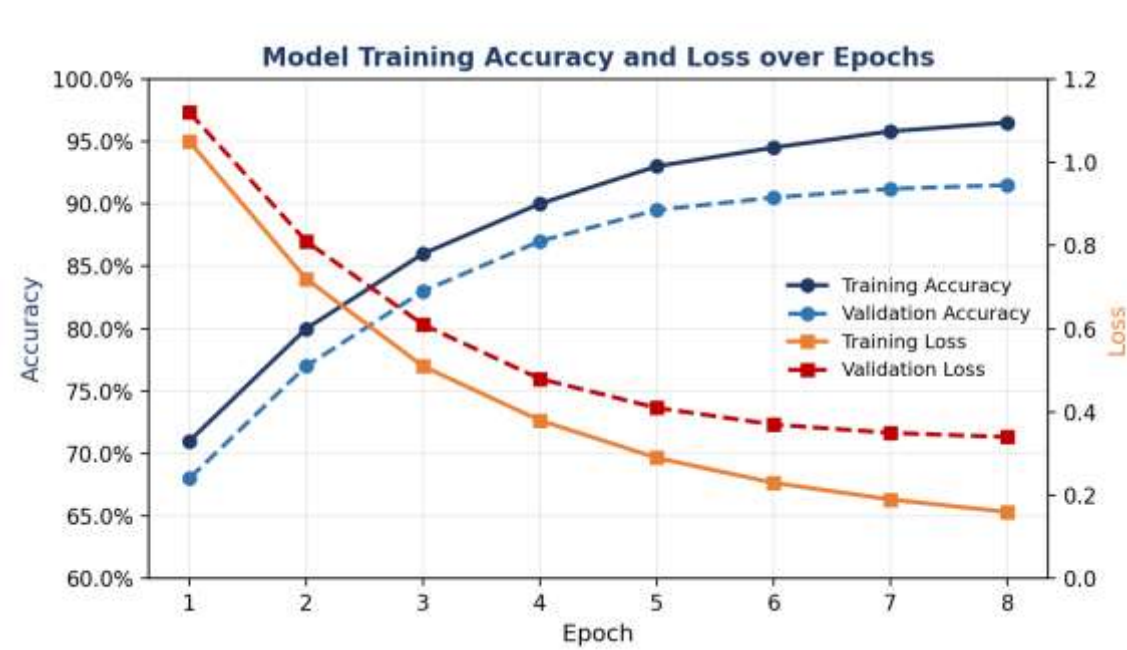
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Deploy Automatically

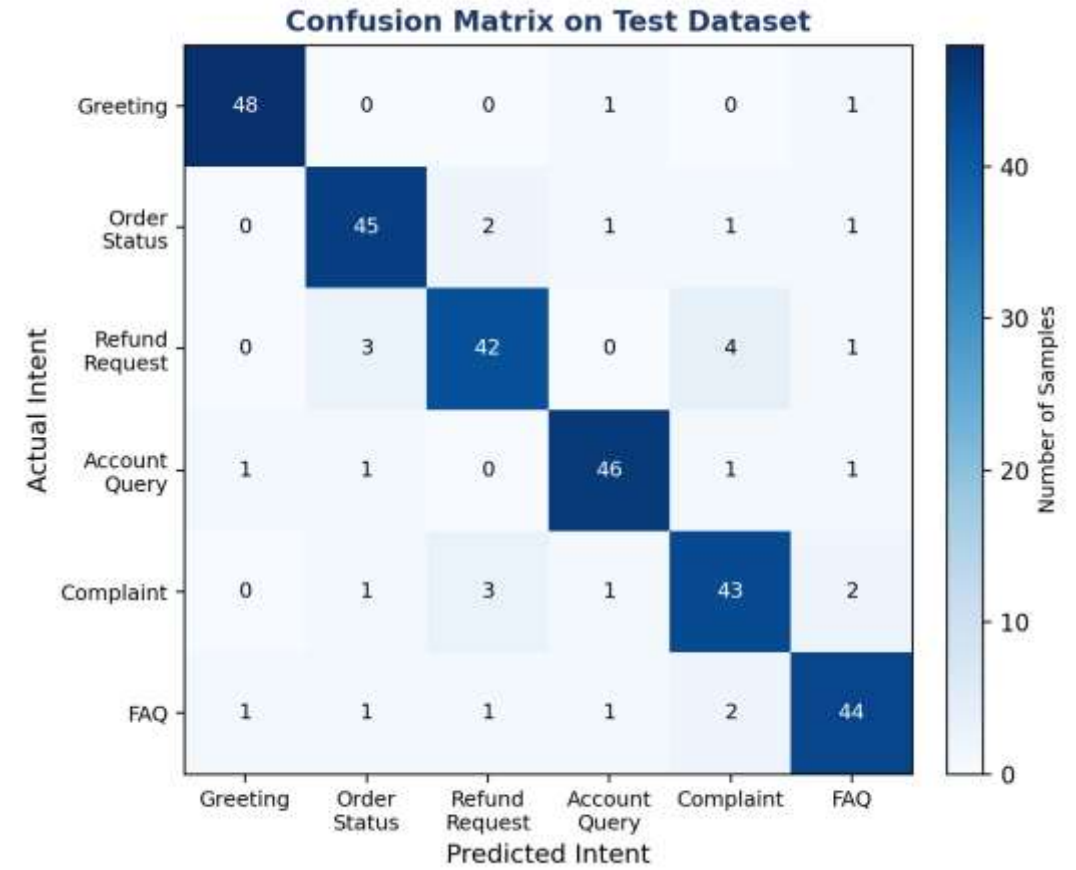
A webhook updates the live application on AWS EC2 / Render.



Results: Model Performance

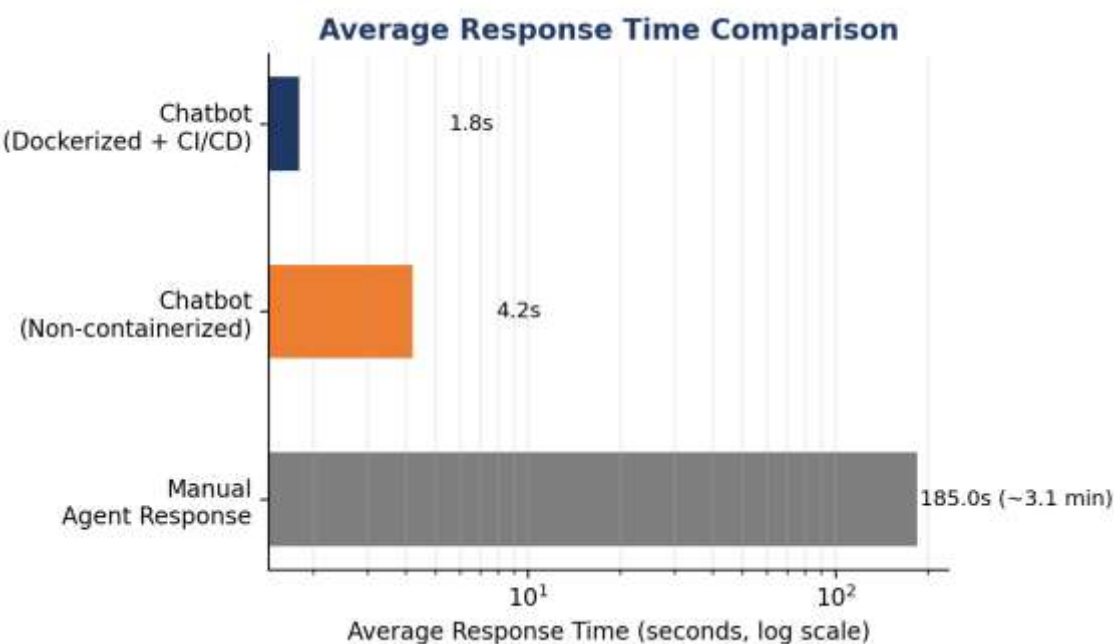
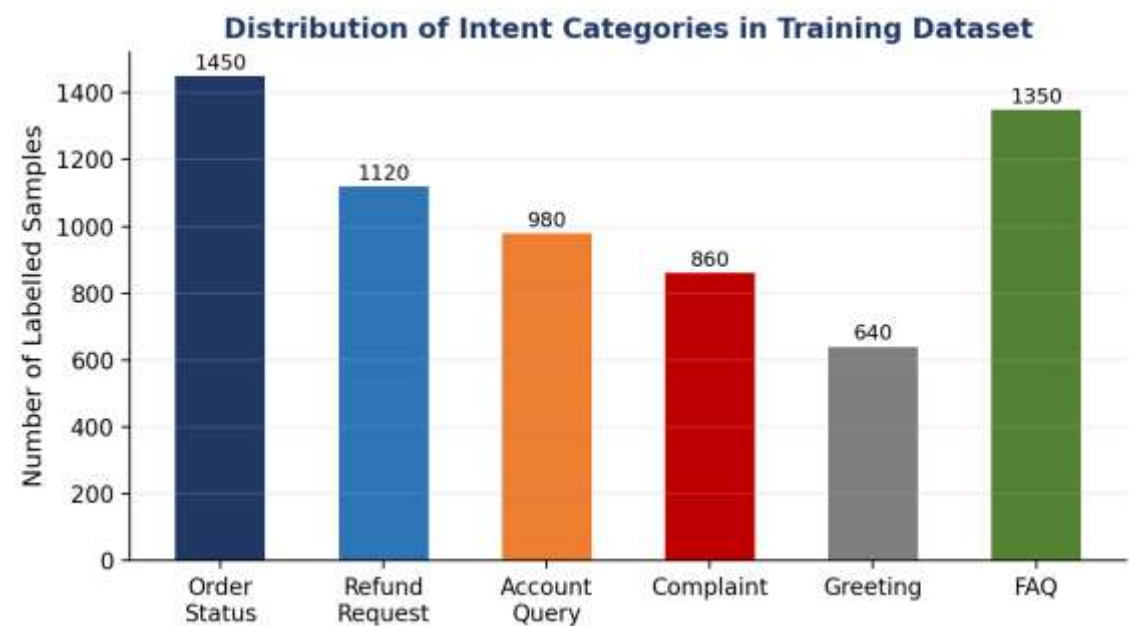


Validation accuracy stabilises around 91-92% by epoch 8 with a small train/validation gap — indicating good generalization without overfitting.



Refund Request and Complaint show the most cross-confusion, reflecting overlapping vocabulary — a target area for future data augmentation.

Results: Deployment Performance



91-92%

Test Accuracy

1.8s

Avg. Response Time
(Dockerized + CI/CD)

100x+

Faster than Manual
Agent Response

8 wks

End-to-End Build
& Deployment

Key Outcomes Achieved



Fine-tuned DistilBERT model classifying customer intent with ~91-92% test accuracy



Low-latency FastAPI REST endpoint (/predict) returning intent + confidence score



Portable, containerized application deployable consistently via Docker



Fully automated CI/CD pipeline — test, build & deploy on every push



Live, cloud-hosted chatbot API accessible over the internet



Reduced response time and operational load on human support agents

Limitations & Future Scope

CURRENT LIMITATIONS

- No multi-turn conversational context
- Fixed set of predefined intent categories
- Unseen intents may be misclassified

PLANNED ENHANCEMENTS

- Multi-turn dialogue management
- Generative responses via LLM integration
- Active learning loop for continuous improvement
- Kubernetes autoscaling + monitoring (Prometheus/Grafana)
- Multilingual intent support

Internship Completion Certificate





Thank You